CHAGEE

CHAGEE SINGAPORE

THE OPPORTUNI-TEA — UNLOCKING GROWTH AT CHAGEE SG WITH AI

OPIM639 · GROUP 3 · KEZIAH, JIN, AYUMI, REVAN · OCTOBER 2026



AGENDA

Spilling the Tea

01 **Introduction**
Chagee Singapore, our scope and why

02 **Executive Summary and Business Plan**
Two key challenges and one strategic decision

03–08 **Our Business Recommendations**
Dynamic demand management · AI-enabled in-store planning · AI-enabled new-store site selection

09 **Execution Plan**
How can we operationalise this business plan

10 **Business Impact**
Financial and non-financial benefits of the business plan

11 **Q & A**

\+ **Appendix**
Supporting evidence



Today, we're presenting Chagee

A tea brand most of you know: order in the app, collect at the counter, or have it delivered to your location.



Our scope

Chagee Singapore's management and its customers

Outlets are **company-owned, not franchised**, so the counter, roster and equipment are Chagee's own calls.

Why Chagee?

1. **Many stores, managed by one company**, one data system, with lots of data
2. **App-centric**: collects personal and transactional data (their rich data is our oil)
3. **Fast-growing, playful**: rapid expansion, room for improvements with AI
4. **An insider's view**: we were fortunate to interview a staff member (anonymous)

FROM FARM TO CUP · OUR PROJECT FOCUS IS DOWNSTREAM

Source
Yunnan tea; local milk and fruit

Warehouse
Restocking from forecasts

Upstream data and decisions sit with **Chagee corporate**

Outlets
Make and hand over

via Chagee App

Last mile
Collection or delivery

Downstream data is analysed and decisions are made by **Chagee SG** (APAC HQ)

OUR FOCUS | The downstream operational challenges: some of the most pressing issues for Chagee Singapore's management.

Our business plan to solve Chagee's problem statements



CASE STUDY · OUR INTERVIEW

"Mei"

Membership Operations · APAC, Chagee

Persona: name changed, interviewee anonymised

WHAT MEI TOLD US

Through the interview, we learnt about Chagee's **business model**, its **local operations**, and some of its most **pressing concerns and strategic needs**, which AI could potentially help solve.

TWO KEY CHALLENGES + ONE STRATEGIC DECISION POINT

KEY CHALLENGE

Online retailing

Use data for better demand management, to grow sales and manage the customer experience.

PROPOSAL 1

Dynamic demand management

At peak, AI reroutes pickups and deliveries to nearby outlets with room, decreasing drop-offs and wait times, and increasing overall sales.

See how it works →



DECISION-MAKER

Zone and branch manager

KEY CHALLENGE

In-store management

A faster, smoother counter: shorter waits, a crew matched to demand and less wastage.

PROPOSAL 2

AI-enabled in-store planning

Higher-precision outlet and zone forecasts help managers decide on staffing, tea machines and fresh stock to maximise store performance.

See how it works →



DECISION-MAKER

Zone and branch manager

STRATEGIC EXPANSION DECISION

New store openings

Where to open next, so a new store is profitable and adds customers instead of moving them.

PROPOSAL 3

AI-enabled new-store site selection

Picks new sites that add customers and profit.

See how it works →



DECISION-MAKER

Expansion and growth manager

Proposal 1: Dynamic demand management

KEY CHALLENGE 1

Demand management: crowded outlets lose sales, with impacted customer experience



WHAT IT COSTS CHAGEE TODAY

Drop-offs
Customers see the wait and leave without ordering.

Crowding
Those who stay crowd the outlet, so it sells less and delivery orders slow down.

Long waits
Customers wait longer than they expected; the experience suffers.

Delivery orders arrive blind
Delivery customers cannot see the in-store queue, so orders keep coming and riders wait at the counter.

Spoils the in-store image
Chagee's elegant stores are part of the brand. A crowd of frustrated customers at the door undoes the classy experience the brick-and-mortar model is built on.

With AI, this can change

- **Higher sales**
  Demand spread across nearby outlets instead of lost.
- **A better brand image**
  Faster turnaround, well managed, a smart system.
- **Calmer counters**
  Delivery riders do not wait long or crowd the counter.

From Chagee's data to demand spread across outlets

An AI agent uses predictive AI to recommend, generative AI explains, and people make the final call.

1 The data

What Chagee collects, or could collect

Chagee app & sales
Orders by outlet and time · Customer location · Margin per drink

Outlet operations
Live queue of open orders · Crew on shift · # of machines

Delivery apps
Delivery orders · Rider waits

Historical events
Decisions made · Profitability outcomes

2 Proposed AI agent

Trigger, recommend, decide, with guardrails

TRIGGER

Wait runs long
A store's wait is, or is about to go, above 40 min.

AI AGENT

Recommend outlet, action, incentive
Predictive AI on live store data and patterns learnt from past decisions, within approved rules.

PEOPLE DECIDE

Customers and managers choose
Generative AI explains each recommendation; they accept or decline.

3 What it does

Example of actions at peak, and it keeps learning

Offers a faster outlet
Routes pickups to the outlet that can hand over sooner.

Recommends a delivery pause
Recommends pausing third-party delivery orders and redirecting demand; the zone manager reviews and approves.

CONTINUOUS LEARNING

Gets smarter over time
Every new decision and its outcome feed back in, so future decisions keep improving.

THE OPERATIONS IDEAS BEHIND IT

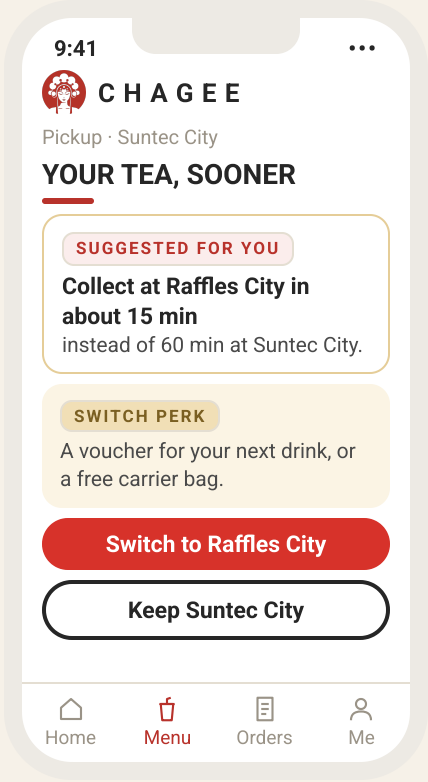
Demand shaping	moving demand strategically
Capacity pooling	with nearby outlets
Reducing bottlenecks	dynamically

KEY BENEFITS

- ✓ **Balances the zone**, thereby increasing sales
- ✓ **Learns from past decisions and live data from stores** to raise profit while enhancing customer experience

Illustration of Proposal 1: Direct customers to outlets with shorter waiting times.

PICKUP · CHAGEE APP



A faster outlet, offered

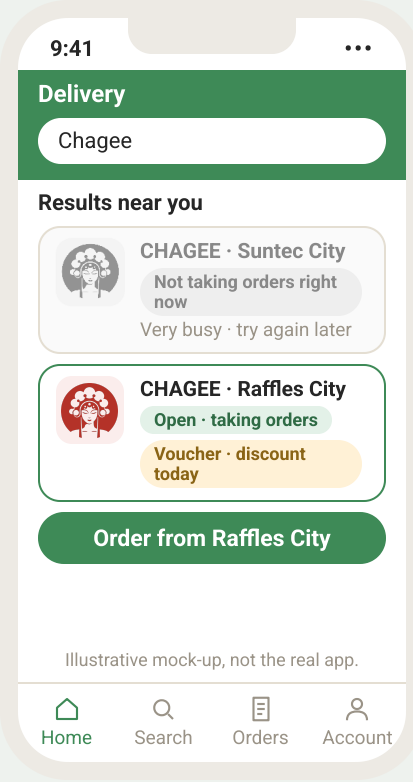
Suntec City is swamped, so the app suggests Raffles City and explains why.

45 min sooner
collect in about 15 min instead of 60, walk included

DYNAMIC PRICING
A voucher or a carrier bag nudges and rewards the switch, offered only while Suntec City is swamped.

Customer chooses before paying.

DELIVERY · E.G. GRABFOOD



The busy outlet steps back

The agent recommends pausing Suntec City on delivery apps and redirecting demand; the zone manager reviews and approves.

Orders flow to Raffles City
no riders stuck at the busy counter

DYNAMIC PRICING
Instead of closing Suntec City on delivery altogether, a voucher at Raffles City can draw demand to the outlet with room.

Zone manager approves the pause and redirection. Placed orders stay put; platform permissions to confirm.

Proposal 2: AI-enabled in-store planning to achieve high-performing outlets

KEY CHALLENGE 2

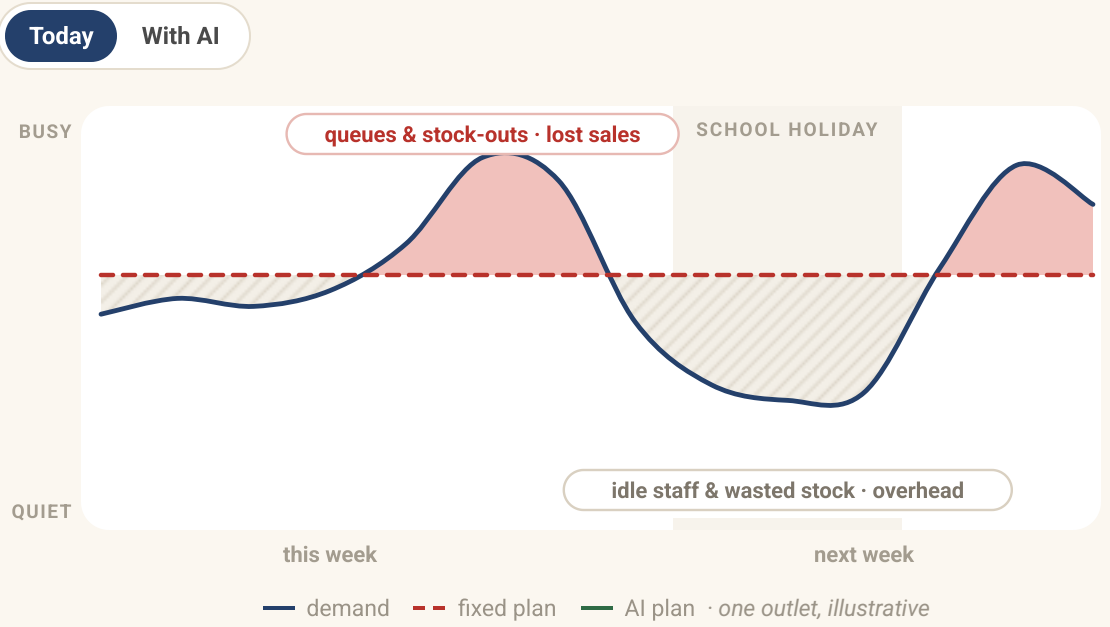
In-store management: matching staff, machines and stock to daily demand

IN PLACE TODAY *Chagee annual report, FY 2025*

- Group-level demand forecasting for stock and inventory
- Centralised, digital store management

ENVISIONED MODEL *enhanced with predictive analytics and AI*

- Forecasts by outlet per day
- AI-recommended rosters, staff moves and machines to run
- Management of fresh inventory, less waste
- Unoptimal store set-ups flagged for review



Proposal 2: AI-enabled in-store planning to achieve high-performing outlets

KEY CHALLENGE 2

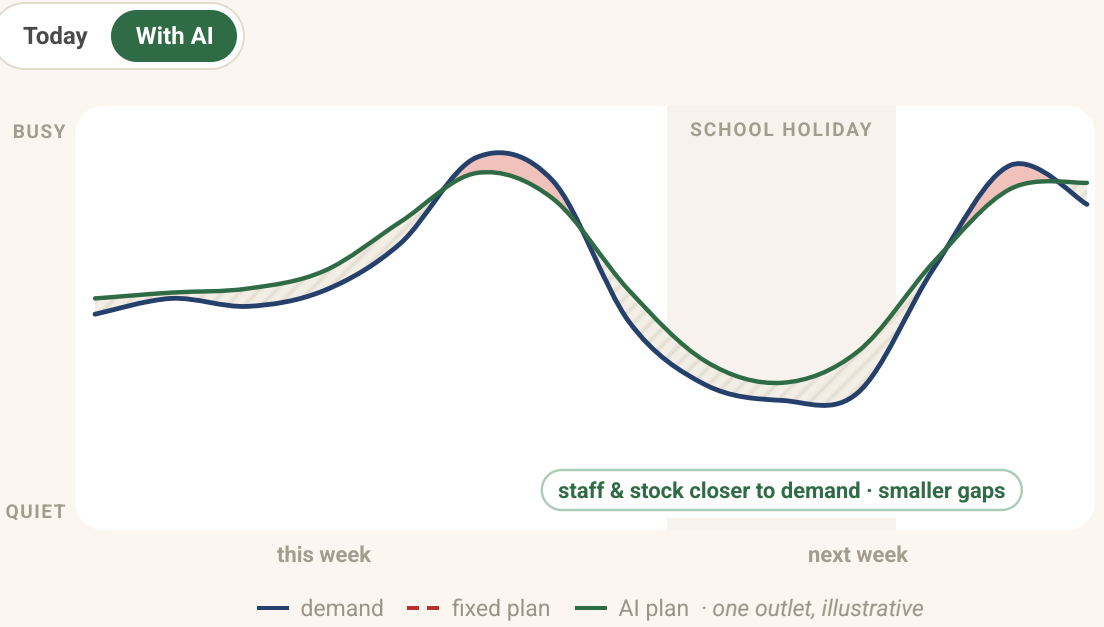
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With AI, this can change

- **Fewer lost sales** — Ready for busy days.
- **Less idle staff** — Rosters flex on quiet days.
- **Less wasted stock** — Fresh prep matches demand.
- **More optimal machinery count** — How many to match demand.

Managers can use insights to plan and manage stores better

HOW IT WORKS

1. **DATA** — examples: orders · staffing · machines
2. **PREDICTIVE AI** — forecasts each outlet by day
3. **AI PLATFORM** — provides recommendations and a chatbot
4. **MANAGERS** — view insights and plan the week

Chagee Store Insights Dashboard – Zone View · Updated 7am daily

STAFF FOR THE COMING WEEK

OUTLET	RECOMMENDATION
Raffles Place	Reassign 1 crew → City Hall
Suntec City	Maintain current crew
City Hall	Receive 1 crew from Raffles Place

WHY School-holiday pattern learnt from past years: office areas like Raffles Place quieten, leisure areas like City Hall get busier.

UPCOMING PEAKS & QUIET SPELLS

▲ Peak	Suntec City · Fri to Sun, afternoons
▲ Peak	City Hall · weekdays, school holiday
▼ Quiet	Raffles Place · Mon to Fri, school holiday

TEA MACHINES · IN STORE

	CURRENT	RECOMMENDED
Raffles Place	4	3 ▼
Suntec City	4	4 =
City Hall	3	4 ▲

FRESH STOCK · PACKAGES PER WEEK

	CURRENT	RECOMMENDED
Raffles Place	9	7 ▼
Suntec City	8	9 ▲
City Hall	6	8 ▲

One model that keeps getting smarter

1. **Events feed in** — Rerouted pickups, delivery pauses and vouchers from Proposal 1
2. **The model retrains** — on these events and the demand they shifted
3. **Forecasts sharpen** — so this dashboard anticipates better, even after protocol changes

Over time, Chagee manages its stores better, ahead of demand rather than reacting to it.

THE OPERATIONS IDEAS BEHIND IT

- **Demand forecasting** — by outlet and day
- **Capacity planning** — staff, machines and stock to the forecast
- **Throughput** — find and fix the slowest step
- **Staff cross-training** — crew cover different roles / outlets

Proposal 3: AI-enabled new-store site selection

BUSINESS PROBLEM A new outlet may look successful while mostly moving sales from nearby stores, while other customers have no convenient outlet. **Context** Chagee aims to expand rapidly in Singapore; the number of stores is confidential (Chagee source, anonymous). **Simulated** searches below.

1 WHO

Where should we open next?



Which location brings us new customers?
HEAD OF EXPANSION

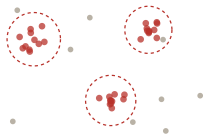
- **A** **Beside an existing outlet**: its sales may come from the store next door.
- **B** **Near underserved customers**: people searching with no outlet nearby.

The manager needs to compare locations, customer access and potential sales—not rely on intuition alone.

2 WHAT

Bring the signals together

- **App searches and unmet demand** — IN THIS DEMO · SYNTHETIC
- **Existing outlets and historical orders** — IN THIS DEMO
- **Residents, workers and transit access** — PROPOSED · NEEDS DATA ACCESS
- **Footfall and nearby competitors** — PROPOSED · NEEDS DATA ACCESS
- **Area delivery demand, if available** — PROPOSED · NEEDS DATA ACCESS
- **Rent and operating costs** — ASSUMED VALUES · P&L VIEW



DBSCAN: unsupervised machine learning that finds dense groups of searches.

These clusters suggest locations to investigate. A separate demand model estimates sales and how much would shift from nearby outlets.

3 RESULT

Compare the options. Make the call.

This site looks promising. Let's compare the assumptions and downside before we commit.
HEAD OF EXPANSION



Site comparison synthetic app data · our assumptions

NET ADDITIONAL CUPS A DAY FOR CHAGEE

B **+731** cluster near Choa Chu Kang

A **+216** beside 34 · West Mall

SALES MOVED FROM NEARBY OUTLETS

B 106 of the 837 cups it sells · **A** 64% of its 604

DEMAND, DELIVERY AND COMPETITION hand-set scores, illustrative pin (appendix)

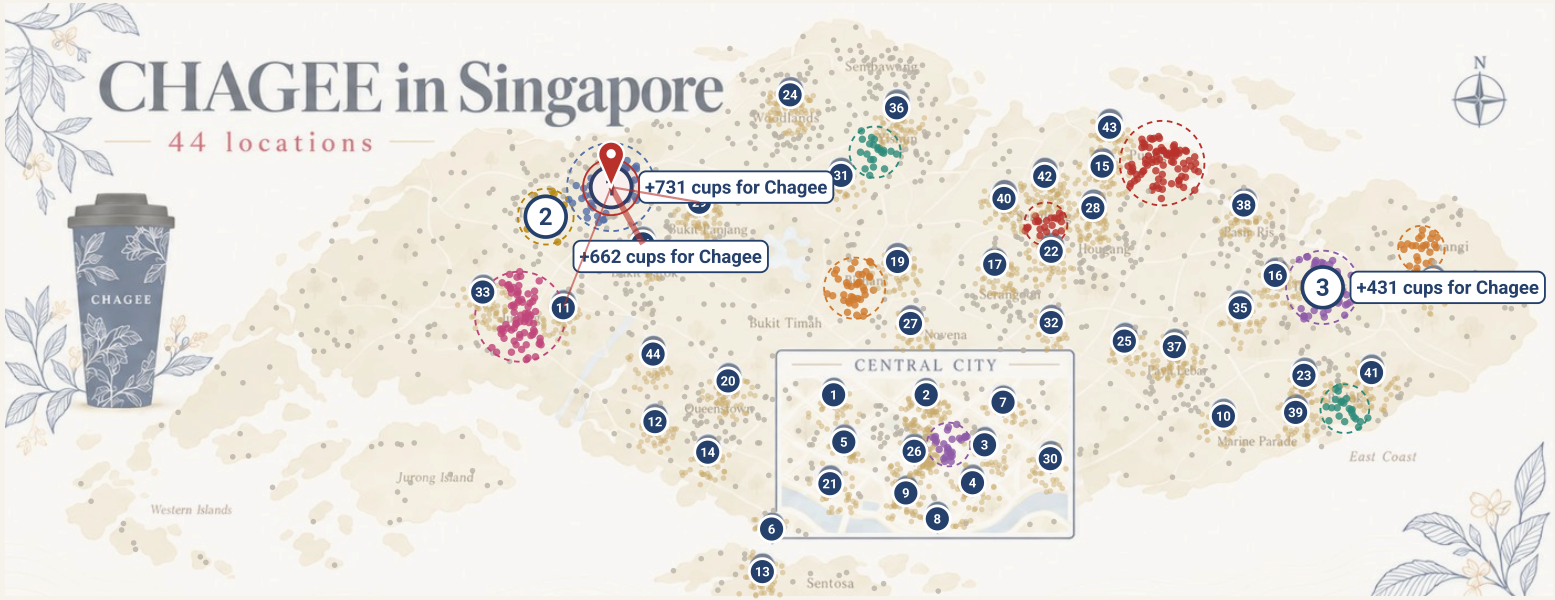
Demand	Delivery	Competition

An AI-assisted dashboard helps the team make an evidence-informed decision. The manager makes the final call.

Proposal 3: AI-enabled new-store site selection

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NEW-STORE FORECAST



SUGGESTED SITE 1 · 2.7 KM FROM 34 · WEST MALL

Strong site: mostly new cups

+731 cups a day for Chagee

The new store sells 837 a day (1.5× an average outlet). 106 of those move from outlets nearby:

34 · West Mall, 2.7 km	**−77**
29 · Bukit Panjang Plaza, 3.3 km	**−28**

Peak hour 13:00: **92 cups** → **2 people** on today's counter (counter pages, appendix).

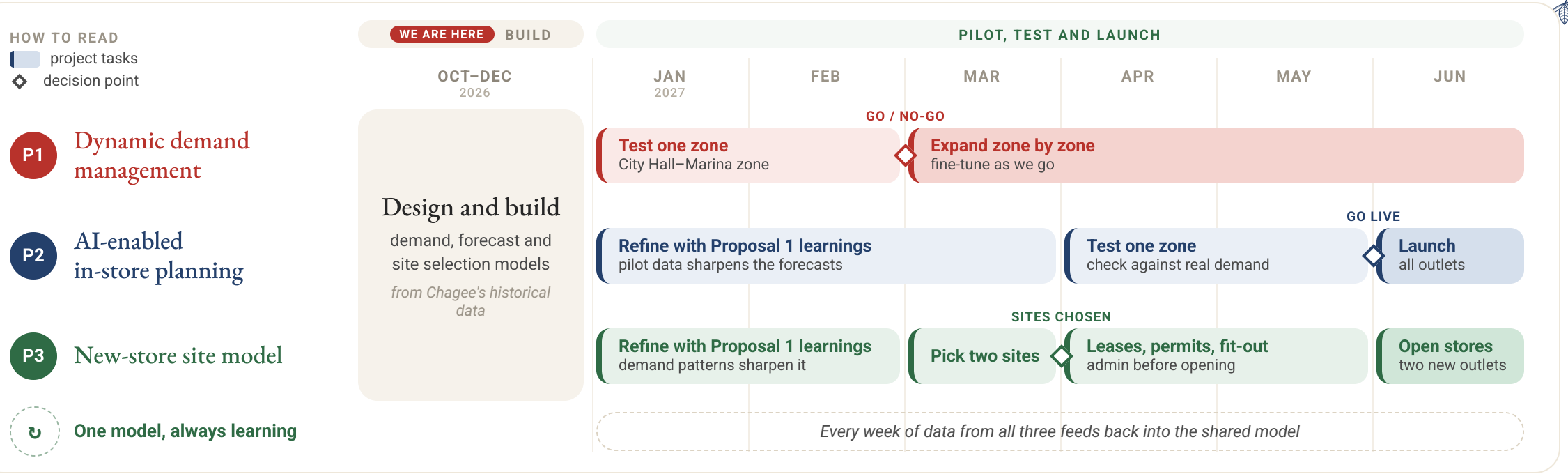
Synthetic app data on the team's map; outlet positions approximate (the central inset is enlarged). Not a trained model: the point is the method. Its five-year P&L: appendix.

TAKEAWAY | Our model picks the site that adds the most cups to Chagee: the cluster near Choa Chu Kang, about **+731** a day. Judged on sales alone, the same store beside an existing outlet looks busy (about 604 cups a day) but adds only **+216**: the rest moves from the outlet next door.

EXECUTION PLAN

Three months to plan, then a phased six-month launch

One integrated AI solution: plan from now to December, start small in January, learn, then scale.



PROJECT TEAM — Our team: Project manager, AI engineer + From Chagee: Supply chain manager (APAC HQ), Zone manager, Store manager

By end June 2027

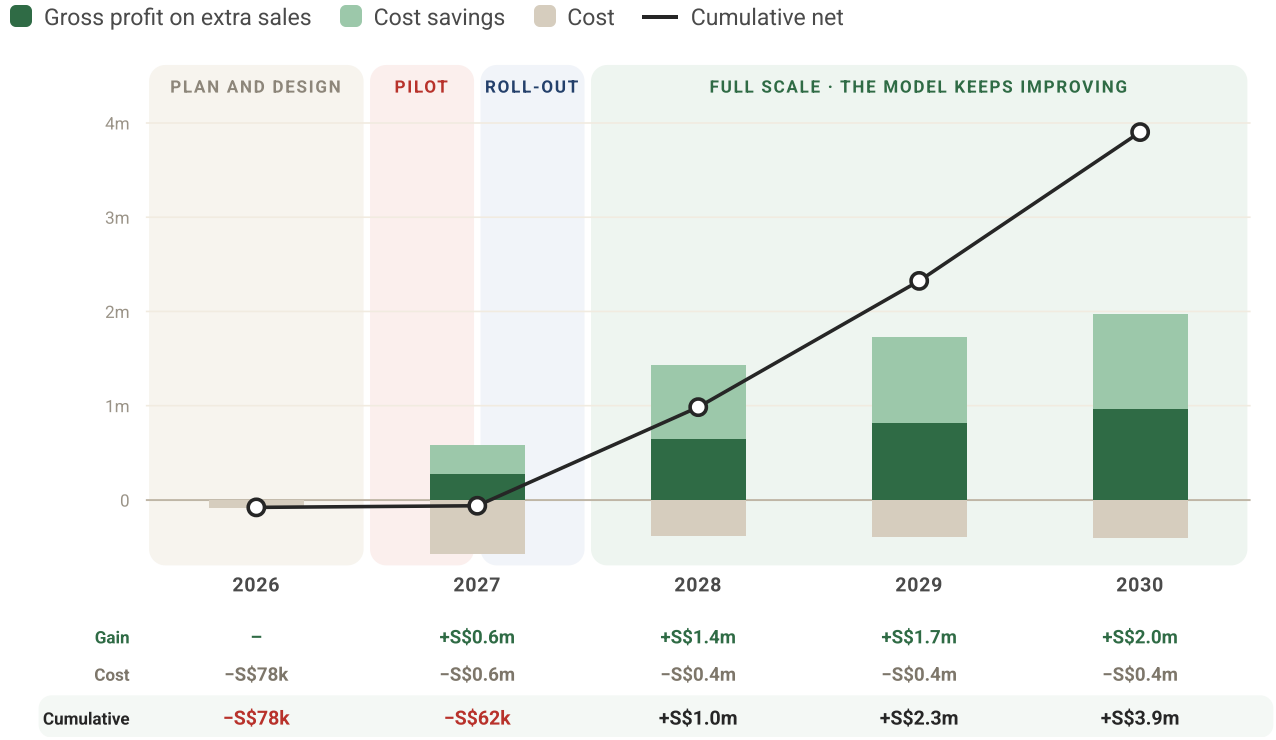
- ✓ **One integrated platform**, with Proposals 1, 2 and 3 launched across Singapore, improving demand management, decision making and site selection
- ✓ **Two new stores** opened in strong Singapore locations, chosen to maximise profit

FINANCIAL IMPACT · SINGAPORE · FIVE YEARS

We forecast our business plan to add S$3.9m net profit over five years and break even in early 2028

Team set up in 2026, pilots in early 2027, expands outlet coverage from 2H-2027 onwards; then the model keeps improving.

Financial gain and cost each year



	2026	2027	2028	2029	2030
Gain	–	+S$0.6m	+S$1.4m	+S$1.7m	+S$2.0m
Cost	−S$78k	−S$0.6m	−S$0.4m	−S$0.4m	−S$0.4m
Cumulative	−S$78k	−S$62k	+S$1.0m	+S$2.3m	+S$3.9m

REVENUE IMPACT

+S$5.0m extra revenue over five years

About 305,000 more cups a year by 2030

P1	Fewer walk-aways at peak	S$2.1m
P2	Staff and machines ready for the peak	S$1.1m
P3	Better-placed new stores	S$1.8m

COST IMPACT

S$3.0m saved

Less idle staff time, fresh-stock waste and machines

S$1.8m invested

S$0.6m one-off, then S$0.4m a year

FINANCIAL RESULT

S$3.9m net over five years

Breakeven **early 2028**

S$1.6m a year at full scale

Full financial model →

BEYOND THE NUMBERS — Brand value · A better customer experience · Calmer, better-looking stores

Harder to measure, just as important

THANK YOU

QUESTIONS AND DISCUSSION

OPIM639 · GROUP 3 · KEZIAH, JIN, AYUMI, REVAN



APPENDIX

SUPPORTING DETAILS FOR THE PRESENTATION

FINANCIALS & SUPPORTING ILLUSTRATIONS FOR PROPOSALS

The five-year model behind the financial impact

S$ THOUSANDS UNLESS STATED	2026	2027	2028	2029	2030	TOTAL
REVENUE IMPACT						
Proposal 1 · cups recovered from walk-aways (thousands)	–	42	91	107	118	357
Proposal 2 · extra cups served at the peak (thousands)	–	20	50	59	66	195
Proposal 3 · extra cups at better-chosen sites (thousands)	–	25	64	92	122	302
Additional revenue	**–**	**513**	**1,207**	**1,519**	**1,799**	**5,038**
Less ingredients and packaging	–	(236)	(555)	(699)	(828)	(2,317)
Gross profit from additional sales	**–**	**277**	**652**	**820**	**972**	**2,720**
COST IMPACT · LEANER OPERATIONS (PROPOSAL 2)						
Staff overhead saved	–	130	328	386	427	1,272
Fresh stock waste avoided	–	141	356	419	464	1,380
Tea machines avoided	–	37	91	103	111	342
Operating cost savings	**–**	**307**	**775**	**908**	**1,002**	**2,993**
Total benefit	**–**	**584**	**1,427**	**1,729**	**1,974**	**5,714**
ADDITIONAL COST FROM INITIATIVES						
Project team	78	312	–	–	–	390
Platform build	–	100	–	–	–	100
Data pipelines and cloud set-up	–	60	–	–	–	60
Security and PDPA review	–	20	–	–	–	20
Run team	–	–	240	240	240	720
Software maintenance	–	–	20	20	20	60
Cloud, licences and AI services	–	52	70	70	70	262
Switch perks (Proposal 1)	–	23	50	59	66	198
Total cost	**78**	**568**	**380**	**389**	**396**	**1,811**
Net financial impact	**(78)**	**16**	**1,047**	**1,340**	**1,578**	**3,903**
Cumulative net impact	*(78)*	*(62)*	*985*	*2,324*	*3,903*	

Key assumptions

Some based on Chagee's real online data; the rest are illustrative assumptions

BASELINE

44 outlets × 240,000 cups a year at S$5.90, growing 3% a year, plus the new outlets: the base the percentages below apply to. Gross margin 54%; wages rise 3% a year

ROLL-OUT

Pilots in the first half of 2027; every outlet from the second half. The model reaches its full effect by 2030

PROPOSAL 1

3% of cups lost at peak, 30% of them recovered; 2% of orders redirected, 50% of them with a S$0.50 perk

PROPOSAL 2

10 idle staff hours a week saved per outlet; fresh-stock waste of 3% of sales cut by 20%; 0.5% more cups at the peak

PROPOSAL 3

Two new outlets in 2027, then one a year, each selling 10% more than a conventional site pick

COSTS

Platform build S$100k, data set-up S$60k and security review S$20k in 2027; then a run team S$240k, cloud S$70k and maintenance at 20% of the build a year. Chagee's managers and store teams are existing staff: no extra cost

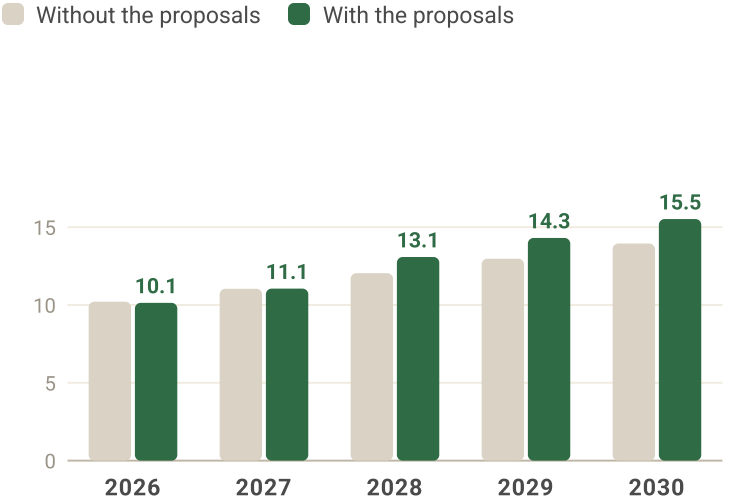
With the proposals, Chagee Singapore profit grows to S$15.5m by 2030 with a 19.6% margin

The whole Singapore business, growing 3% a year with new outlets (49 by 2030), with and without the proposals.

S$ MILLIONS UNLESS STATED	2026	2027	2028	2029	2030	TOTAL
Outlets at year end	44	46	47	48	49	
Cups sold (millions)	10.6	11.2	12.0	12.7	13.4	59.9
REVENUE						
Baseline sales	62.3	65.6	69.9	73.5	77.3	348.6
Additional sales from the proposals	–	0.5	1.2	1.5	1.8	5.0
Total revenue	**62.3**	**66.1**	**71.1**	**75.0**	**79.1**	**353.6**
Cost of goods sold	(28.7)	(30.3)	(32.3)	(34.1)	(35.9)	(161.3)
Gross profit	**33.6**	**35.9**	**38.7**	**40.9**	**43.2**	**192.3**
Gross margin	*54.0%*	*54.2%*	*54.5%*	*54.6%*	*54.6%*	
OPERATING EXPENSES						
Staff	(8.6)	(8.9)	(9.3)	(9.7)	(10.2)	(46.8)
Rent and utilities	(9.2)	(9.4)	(9.8)	(10.0)	(10.2)	(48.6)
Delivery-app commissions	(5.6)	(5.9)	(6.3)	(6.6)	(7.0)	(31.4)
Equipment: tea machines avoided	–	0.0	0.1	0.1	0.1	0.3
AI platform and team	(0.1)	(0.6)	(0.4)	(0.4)	(0.4)	(1.8)
Total operating expenses	**(23.5)**	**(24.8)**	**(25.6)**	**(26.6)**	**(27.7)**	**(128.2)**
Outlet operating profit	**10.1**	**11.1**	**13.1**	**14.3**	**15.5**	**64.1**
Operating margin	*16.3%*	*16.7%*	*18.4%*	*19.1%*	*19.6%*	

FOR COMPARISON · WITHOUT THE PROPOSALS						
Outlet operating profit without the proposals	10.2	11.0	12.0	13.0	13.9	60.2
Operating margin without the proposals	*16.4%*	*16.8%*	*17.2%*	*17.6%*	*18.0%*	
Uplift from the proposals	**(0.1)**	**0.0**	**1.0**	**1.3**	**1.6**	**3.9**

Outlet operating profit, S$m



REVENUE IN 2030

S$79.1m

up 27% on 2026, with 49 outlets

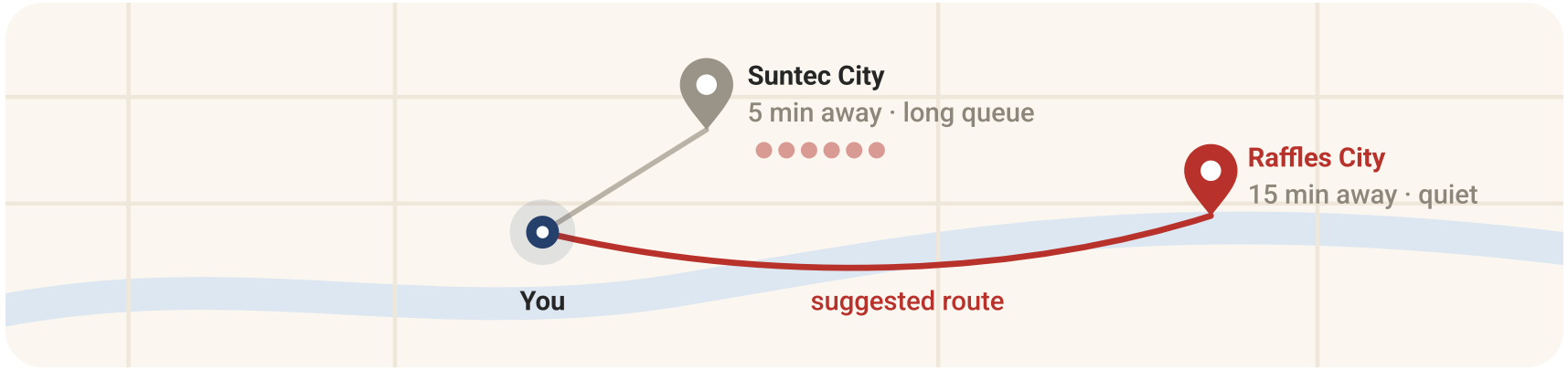
MARGIN IN 2030

19.6%

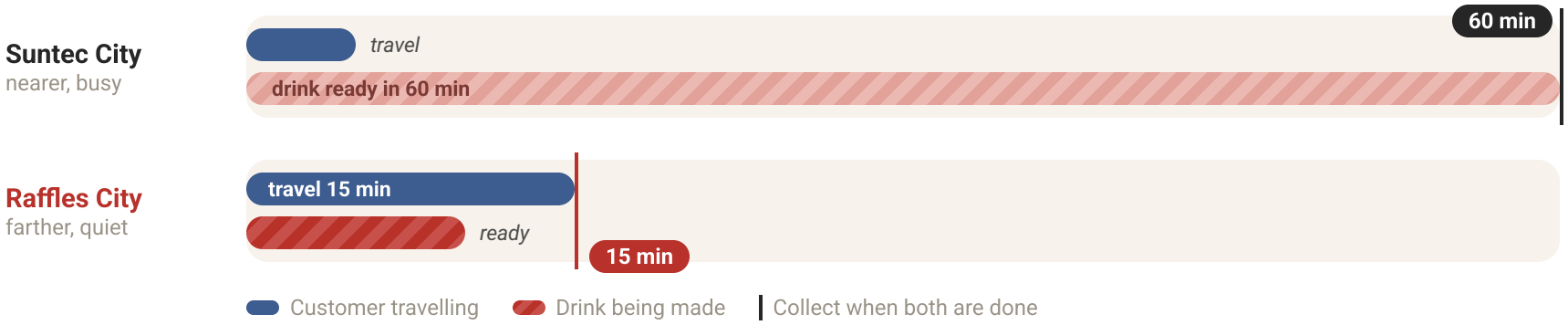
against 18.0% without the proposals

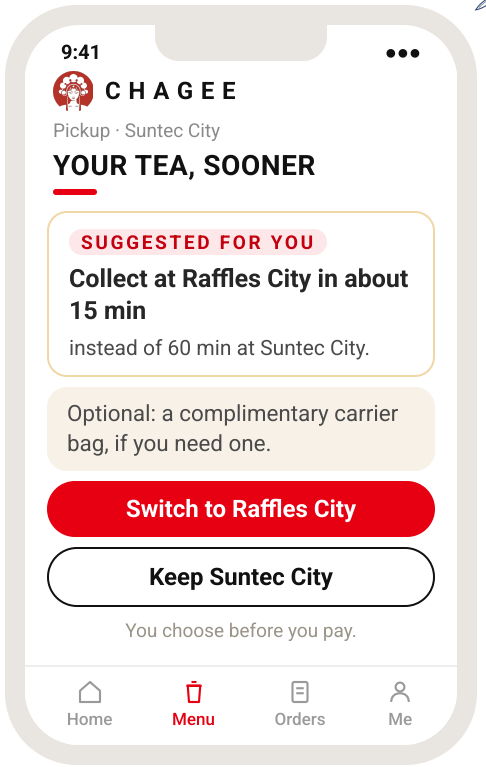
Your tea, sooner: AI routes pickups to a faster outlet

When one outlet is swamped, the app offers a nearby outlet that can hand over the drink sooner, and the customer chooses.



THE RACE TO YOUR DRINK





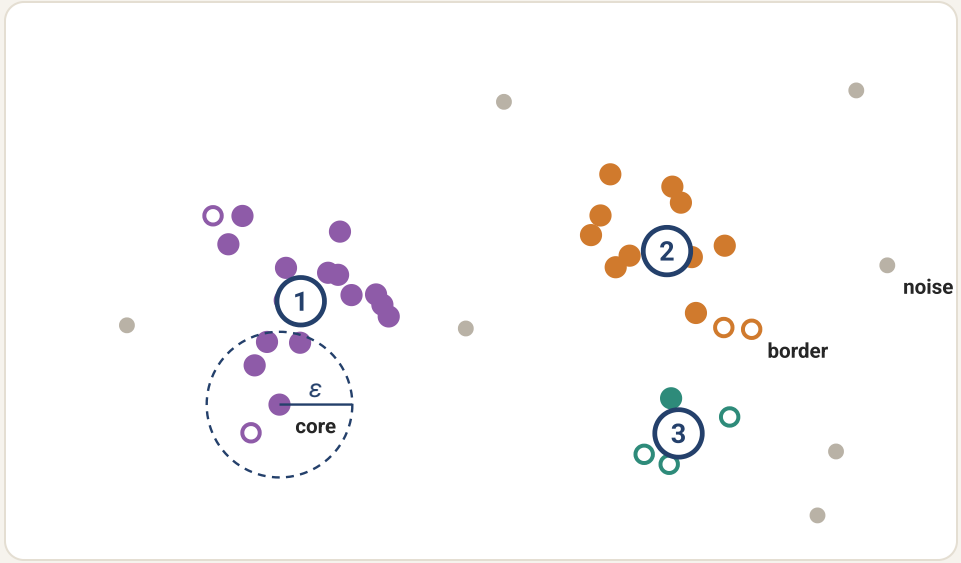
GENERATIVE AI MESSAGE · MOCK-UP

45 minutes sooner

Travel and preparation happen at the same time, so the customer collects as soon as both are done.

DBSCAN finds where searches bunch up; a demand model turns a pin into cups

DBSCAN IN FOUR MOVES · TOY EXAMPLE



1. **Radius ε.** Two searches are neighbours if they are within ε of each other (the cluster-radius slider).
2. **Core.** A search with at least the minimum number of searches within ε (counting itself) is a core point.
3. **Grow.** A cluster spreads from core to core. Searches it reaches that are not core join as border points.
4. **Noise.** Searches no cluster reaches. Unlike k-means, nobody picks the number of clusters, and odd shapes are fine.

Unsupervised: no labels or past openings are needed. Ester, Kriegel, Sander and Xu (1996). On slide 08 it runs only on searches with no outlet within walking distance; each cluster's centre is a candidate site.

THE SIMULATED NUMBERS, IN ORDER · AT THE SLIDER DEFAULTS

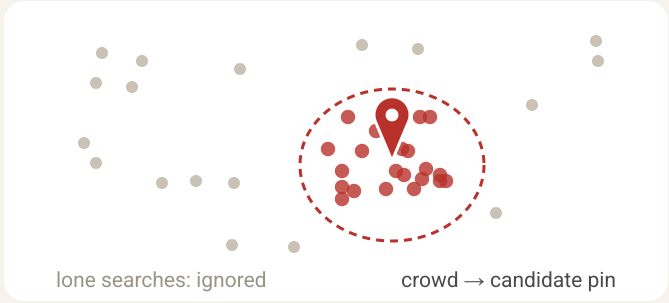
1	**App searches**	Synthetic dots on land: 15% spread evenly, 35% around the outlets' malls, the rest around 20 towns	**3,000 dots**
2	**No store nearby**	The nearest outlet is further than the walking distance	**1 km · 46% of dots**
3	**DBSCAN**	Clusters the no-store-nearby dots	**ε 0.6 km · min 15 · 11 clusters**
4	**Pull of a store**	Fades with distance d: exp(−½ (d/L)²), with the length L set by the walking distance	**L = 1 km**
5	**Who buys where**	A dot buys at a store in proportion to its pull, against the pull of not buying (a store next door = one)	**not buying 0.3**
6	**Calibration**	Each dot's weight is set so the average outlet sells the deck's cups a day (cups a year ÷ open days)	**556 cups a day**
7	**Score a pin**	Cups for Chagee = what the new store sells − what it moves from nearby outlets	**top: 837 − 106 = +731**
8	**Verdict**	Share of the new store's cups that are new to Chagee	**strong ≥ 60% · weak < 40%**
9	**Opening roster**	Peak-hour cups ÷ cups one person handles, with the safety margin	**one per 83 cups**

All synthetic: the dots, towns and weights are assumptions on the team's map (not to scale), not Chagee app logs or population data. With real app logs, steps one and two change; the rest runs unchanged.

DBSCAN picks the spot; a demand model counts the cups

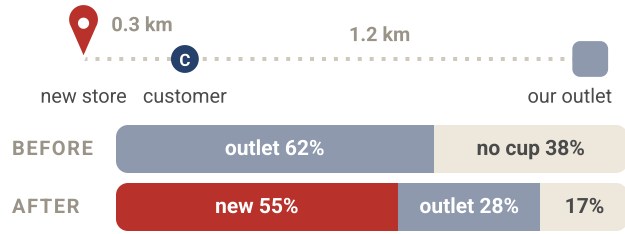
Two separate jobs. DBSCAN never predicts sales: it only finds where unserved searches crowd together.

1 Find the crowd — DBSCAN



Keep the searches with no outlet within 1 km. Ask each one: **are at least 15 searches within 0.6 km of me?** Crowds that touch merge into one; lone searches are ignored. The centre of each crowd is a candidate site.

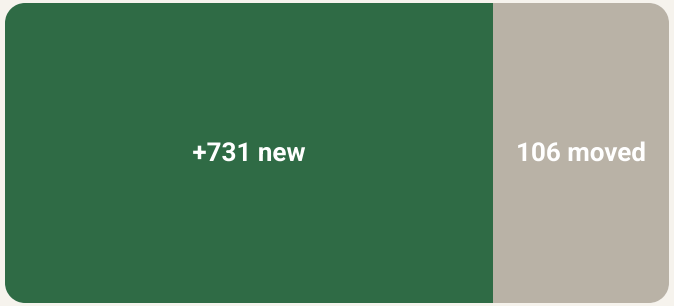
2 Feel the pull — DEMAND MODEL



The closer a store, the more of a customer's cups it gets. Some cups are never bought at all.

When the new store opens, it gets 55% of this customer's cups: **34% taken from our outlet** + **21% never bought before**.

3 Count new, not moved



Add up every customer's share: a new store at the top site sells **837 cups a day**. 106 of them would otherwise have gone to our existing outlets, so **+731 are new to Chagee**.

Strong site if at least 60% of its cups are new; weak below 40%.

IN ONE LINE DBSCAN says **where** to look; the demand model says **how many** cups a store there would sell, and how many of those are really new.